



ENGINEERS LEVEL · CAPSTONE

Innovation Challenge

Student Workbook

Prototype and pitch a tech solution to a real-world problem — code, AI, security, and ethics together.

Name: _____

THEORY

Build Something That Matters

Your capstone integrates the whole journey: pick a real problem (SDG-style), design a solution using your **full toolkit** (Python/JS + **AI** + **cybersecurity/ethics**), build a **prototype**, run a security and ethics check, **iterate** from feedback, and **pitch** its **impact**.

Discuss: which real-world problem matters to you, who does it affect, and how could technology genuinely help?

Frame it

State your challenge and who benefits.

My challenge (problem):

Who benefits, and how:

Code we use:

Python / JS

AI model

security check

for / if

dict / list

pitch

Words we are learning

- **SDG** — a UN Sustainable Development Goal; a real global problem to tackle.
- **Prototype** — a first working build to test and improve.
- **Iteration** — improving from feedback and testing.
- **Pitch** — a short, clear case for your solution's value.
- **Impact** — who it helps and how much.

Name: _____ **Date:** _____

Choose A Challenge

Pick a real problem worth solving. Answer the framing questions.

The problem / SDG I will tackle:

Who is affected, and why does it matter?

Name: _____ Date: _____

Design The Solution

Choose your tools and sketch the system.

Which will I use? (circle)

Python

JavaScript

an AI model

data analysis

a security/privacy feature

sketch your system / data flow

Name: _____ **Date:** _____

Prototype & Security/Ethics Check

Plan the core build, then run a security and ethics check.

Core feature (pseudocode):

```
def main(): ...
```

Security/ethics check (data protected? honest about AI? fair to all users?):

Name: _____ **Date:** _____

Pitch & Present

Test your prototype, then build your pitch.

- ☐ It addresses the real problem I chose
- ☐ It integrates code + AI and/or security/ethics
- ☐ The prototype works (tested & debugged)
- ☐ I can explain its impact clearly

My pitch (problem → solution → impact):

Name: _____ **Date:** _____

PROJECT

Innovation Challenge — Plan

Your capstone: prototype and pitch a tech solution to a community or global (SDG-style) problem, integrating code, AI, and security/ethics. This completes the Engineers level and the K-8 journey.

1. My challenge and who it helps:

2. Tools I'll integrate (circle):

Python

JS

AI model

data

security/ethics

3. Core feature (pseudocode):

```
def main(): ...
```

4. My pitch in one line (problem → solution → impact):